Python Programming Basics

Assignment Solutions & Code Snippets

1. Print "Hello World"

```
print("Hello World")
```

2. Get User Input Using input() Function

```
user_input = input("Enter something: ")
print("You entered:", user_input)
```

3. Print Your Name with Hello Using input() Function

```
name = input("Enter your name: ")
print("Hello", name)
```

4. Print Bio Data

```
print("--- BIO DATA ---")
print("Name: John Doe")
print("Age: 20")
print("Gender: Male")
print("Qualification: B.Tech Computer Science")
print("City: Ahmedabad")
```

5. Take Only Numerical Input

```
# Converts the input string to an integer
num = int(input("Enter a number: "))
print("The numerical input is:", num)
```

6. Take 3 Different Inputs and Display Their Data Type

```
val1 = input("Enter a string: ")
val2 = int(input("Enter an integer: "))
val3 = float(input("Enter a float: "))

print("Type of val1:", type(val1))
print("Type of val2:", type(val2))
print("Type of val3:", type(val3))
```

7. Convert a Number to Floating Point Number

```
num = int(input("Enter an integer: "))
float_num = float(num)
print("Floating point number:", float_num)
```

8. Add, Subtract, Multiply, and Divide 2 Numbers

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
```

9. Print Quotient and Remainder Separately

```
num1 = int(input("Enter dividend: "))
num2 = int(input("Enter divisor: "))

quotient = num1 // num2
remainder = num1 % num2

print("Quotient:", quotient)
print("Remainder:", remainder)
```

10. Print String 3 Times in One Single Print Statement

```
text = input("Enter a string: ")
print(text * 3)
```

11. Calculate Area of a Circle

```
import math

r = float(input("Enter the radius of the circle: "))
area = math.pi * r * r
print("Area of circle:", area)
```

12. Find Area of a Rectangle

```
l = float(input("Enter length: "))
b = float(input("Enter breadth: "))
area = l * b
print("Area of rectangle:", area)
```

13. Find Volume of a Cube / Cuboid

```
l = float(input("Enter length: "))
b = float(input("Enter breadth: "))
h = float(input("Enter height: "))
volume = l * b * h
print("Volume:", volume)
```

14. Find Area of a Triangle

```
l = float(input("Enter base/length: "))
b = float(input("Enter height/breadth: "))
area = (l * b) / 2
print("Area of triangle:", area)
```

15. Calculate Simple Interest

```
P = float(input("Enter Principal amount (P): "))
R = float(input("Enter Rate of interest (R): "))
N = float(input("Enter Number of years (N): "))

SI = (P * R * N) / 100
print("Simple Interest:", SI)
```

16. Convert Meter to Kilometer

```
meters = float(input("Enter distance in meters: "))
kilometers = meters / 1000
print("Distance in kilometers:", kilometers)
```

17. Convert Fahrenheit to Celsius

```
F = float(input("Enter temperature in Fahrenheit: "))
C = (F - 32) / 1.8
print("Temperature in Celsius:", C)
```

18. Convert Celsius to Fahrenheit

```
C = float(input("Enter temperature in Celsius: "))
F = C * (9 / 5) + 32
print("Temperature in Fahrenheit:", F)
```

19. Compare 2 Numbers Without Conditional Statements

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

result = num1 > num2
print(str(result).lower())
```

20. Find Minimum and Maximum of 2 Numbers

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Maximum number:", max(num1, num2))
print("Minimum number:", min(num1, num2))
```

21. Perform Binary "AND" and "OR" Operations

```
num1 = int(input("Enter first integer: "))
num2 = int(input("Enter second integer: "))

print("Bitwise AND operation:", num1 & num2)
print("Bitwise OR operation:", num1 | num2)
```

22. Check if a Number is Even Without Conditional Statements

```
num = int(input("Enter an integer: "))
is_even = (num % 2 == 0)
print(str(is_even).lower())
```